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9702/52

May/June 2017

Candidates answer on the Question Paper.

No Additional Materials are required.

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

- 1 A student is investigating the motion of a small cube on a turntable connected to an electric motor as shown in Fig. 1.1.

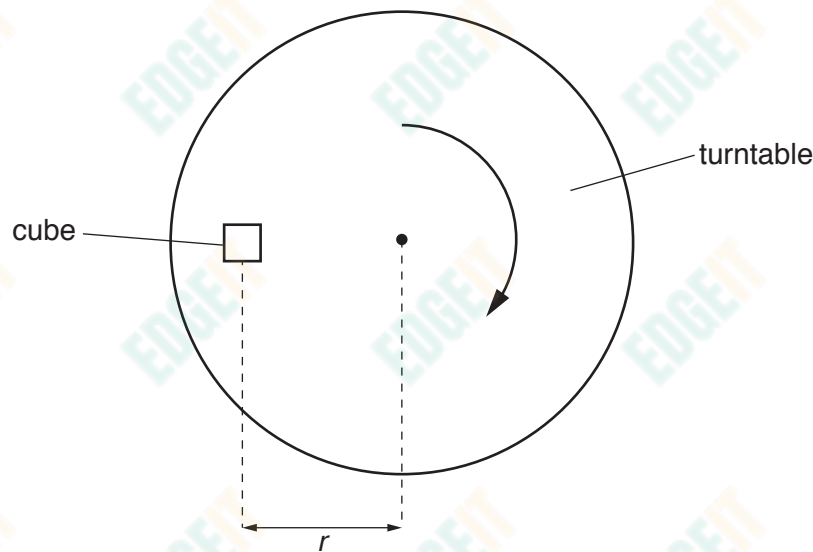


Fig. 1.1

The cube is placed at a distance r from the centre of the turntable. It is suggested that the relationship between r and the maximum frequency f of the turntable for which the cube does not move relative to the turntable is

$$K = 4\pi^2 mfr$$

where m is the mass of the cube and K is a constant.

Design a laboratory experiment to test the relationship between f and r . Explain how your results could be used to determine a value for K . You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

[15]

Diagram

This image shows a blank sheet of white paper with horizontal dotted lines for writing. The lines are evenly spaced and run across the width of the page. There are faint, repeating watermarks of the word "EDGE" in a light blue color, slanted diagonally across the page. The overall appearance is that of a clean, unused piece of stationery or a template for a document.

[Total: 15]

- 2 A student is investigating the current in a circuit. The circuit is set up as shown in Fig. 2.1.

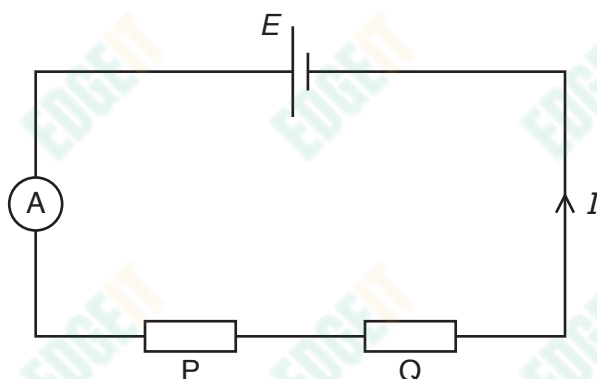


Fig. 2.1

Two resistors P and Q are connected to a power supply of e.m.f. E and negligible internal resistance. The current I is measured.

The resistance of resistor P is P . The experiment is repeated for different values of P .

It is suggested that I and P are related by the equation

$$E = I(P + Q)$$

where Q is the resistance of resistor Q .

- (a) A graph is plotted of $\frac{1}{I}$ on the y -axis against P on the x -axis.

Determine expressions for the gradient and the y -intercept.

gradient =

y -intercept =

[1]

- (b) Values of P and I are given in Fig. 2.2.
The tolerance of each value of P is $\pm 5\%$.

P/Ω	I/mA	$\frac{1}{I}/\text{A}^{-1}$
$180 \pm$	34	
$220 \pm$	28	
$330 \pm$	19	
$470 \pm$	14	
$560 \pm$	12	
$680 \pm$	10	

Fig. 2.2

Calculate and record values of $\frac{1}{I}/\text{A}^{-1}$ in Fig. 2.2.

Determine the absolute uncertainties in P .

[2]

- (c) (i) Plot a graph of $\frac{1}{I}/\text{A}^{-1}$ against P/Ω .

Include error bars for P .

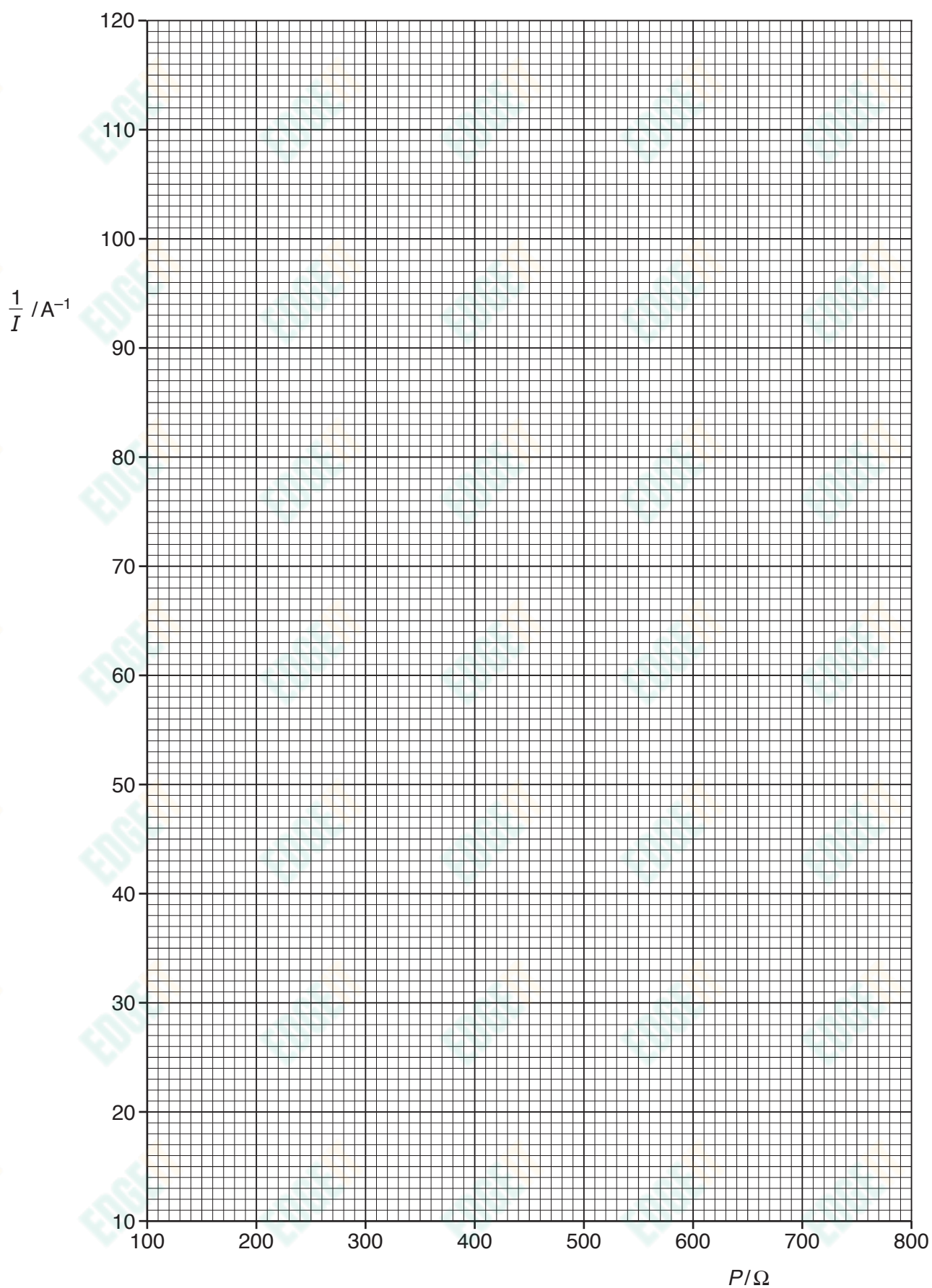
[2]

- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Both lines should be clearly labelled.

[2]

- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of E and Q . Include appropriate units.

E =

Q = [2]

- (ii) Determine the percentage uncertainties in E and Q .

percentage uncertainty in E = %

percentage uncertainty in Q = % [2]

[Total: 15]